environments, and supporting coordinated rescue and relief efforts. Their ability to operate in dangerous or inaccessible areas helps protect rescue personnel, prioritize response actions, and accelerate recovery operations.

Table 8.8 Drone contributions to emergency response and disaster management.⏎

Emergency operation area Search and rescue	**Drone capabilities Thermal**	**Operational impact Locating survivors**	**Outcome/advantage Faster rescue**
Search and rescue missions	Thermal imaging, audio broadcasting, aerial mapping	Locating survivors quickly; guiding rescue teams	Faster rescue operations and reduced risk to rescuers
Disaster impact assessment	High-resolution imaging, LiDAR terrain mapping	Real-time damage estimation and hazard identification	Informed decision-making for relief planning
Wildfire monitoring	Heat signature detection, firefront perimeter tracking	Supports containment strategy; monitors fire spread	Reduced wildfire damage and improved firefighter safety
Flood and cyclone response Pollution and	Water level mapping, route clearance analysis Gas sensors,	Identifies accessible routes for relief supply movement Identifies	Efficient evacuation and supply chain coordination Targeted cleanup and
Pollution and environmental hazard monitoring	Gas sensors, plume tracking, and spill detection	Identifies contamination hotspots and affected zones	Targeted cleanup and reduced ecological harm